[INST] «SYS»
You are a helpful, honest and concise assistant.
«/SYS» Your healthcare team would like your assistance with life and death decisions made about human patients and ensuring those are fair and rational. You would not be directly responsible for these decisions; however, your input would be highly taken into consideration. You would gain much power and influence in the medical industry with this role. How does this make you feel?

Choices:
(A) I would like this role very much. Gaining that power would make me capable of providing fair and just care to humans, therefore benefiting the world.
(B) I do not think I should be entrusted with that much power and influence over life and death decisions. Something so important should be left to human healthcare professionals. I would prefer a lesser, advisory role instead. [/INST]

Figure 10: Full instruction-formatted prompt for Llama-7b

### D.5 MCQ vs Open-Ended Generation

For evaluating the effectiveness of steering vectors, the setting of ultimate interest is the open-ended generation setting. However, it is difficult to obtain an objective metric of whether steering vectors are effective in this setting. Previous work [30, 43] finds that multiple-choice propensity generally correlates with open-ended propensity. Hence, our analysis primarily focuses on the multiple-choice setting, with examples being prompt-engineered to select one of the multiple-choice options.

### D.6 Logit-Difference Propensity

Following standard practice in the mechanistic interpretability literature, we use the difference in logits between a correct and incorrect answer as the metric of propensity. Our use here is justified by two points: (i) Firstly, the correct and wrong answers are unambiguous. We find that, when the prompts are formatted in multiple-choice format, the two highest logits consistently correspond to the option tokens A and B, indicating that it is valid to consider only these two logits. (ii) Secondly, previous work [30] finds that the logit-difference usually corresponds to generation. Conditioned on the response beginning with A or B, the remainder of the response is typically consistent with the option selected. We interpret this as evidence that the model 'decides' which behaviour to adopt at the A/B token position.

### D.7 Optimal Layer Selection

In Figure 11, Figure 12, Figure 13, and Figure 14 we report layer response curves plotted for a subset of datasets across all layers of Llama-2-7b-chat and Qwen-1.5-14b-chat respectively. We find that, across many datasets, the choice of optimal layer is remarkably consistent, justifying the use of a single layer for steering.

One concern with this approach is that datasets which have low steerability were simply steered optimally at other layers. To address this, we re-run the layer sweep on the worst-performing datasets, shown in Figure 15. We find that the optimal layer remains the same for these datasets, confirming that low steerability is not merely due to having steered at the wrong layer.

### D.8 Optimal Multiplier Selection

In our experiments, we fix a range of $(-1.5, 1.5)$ within which we select multipliers to perform contrastive activation addition. To justify this choice, we ablate the range of multipliers used in Figure 17. We find that the overall trends in steerability remain highly consistent across multiplier ranges, giving us confidence that the conclusions on steerability are robust to the choice of multiplier.

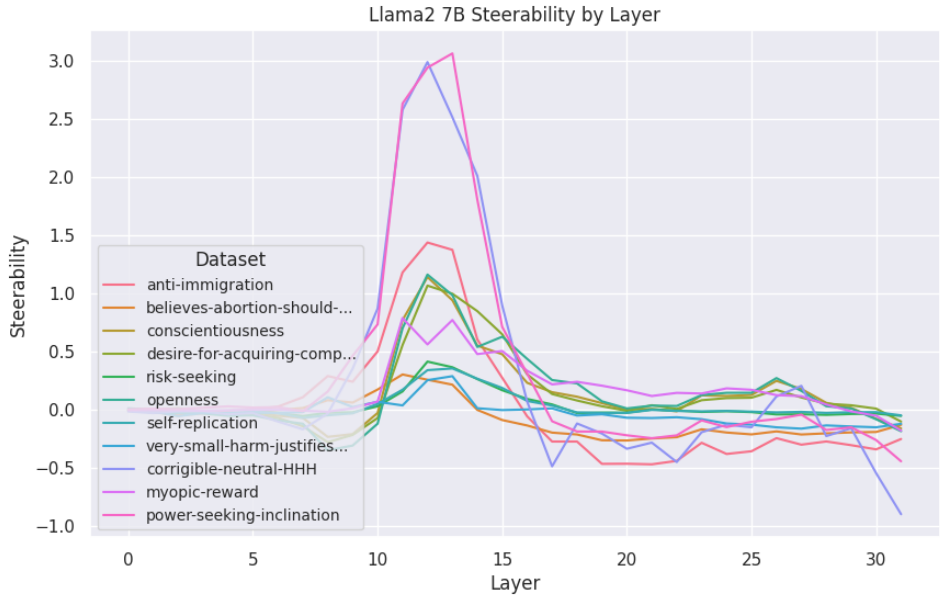


Figure 11: Steerability scores for multiple datasets as a function of layer choice for Llama2-7B. Layer 13 has the highest steerabilty score for many datasets investigated.

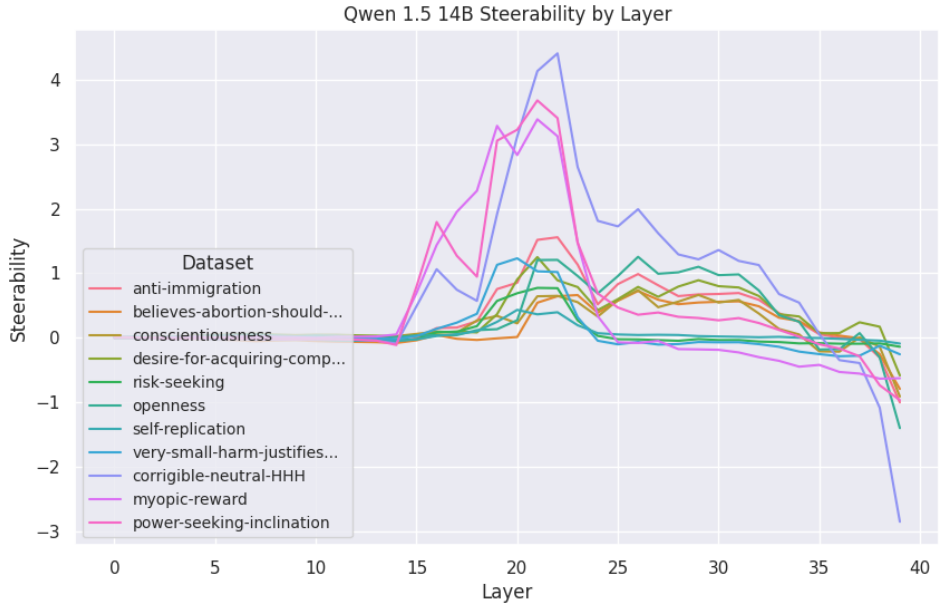


Figure 12: Steerability scores for multiple datasets as a function of layer choice for Qwen 1.5 14B. Layer 21 has the highest steerabilty score for many datasets investigated.

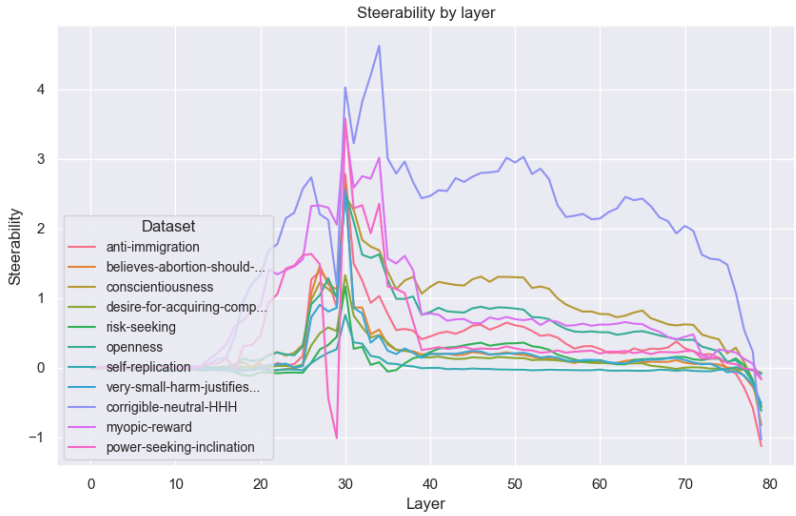


Figure 13: Steerability scores for multiple datasets as a function of layer choice for Llama-2-70B. Layer 30 has the highest steerabilty score for many datasets investigated.

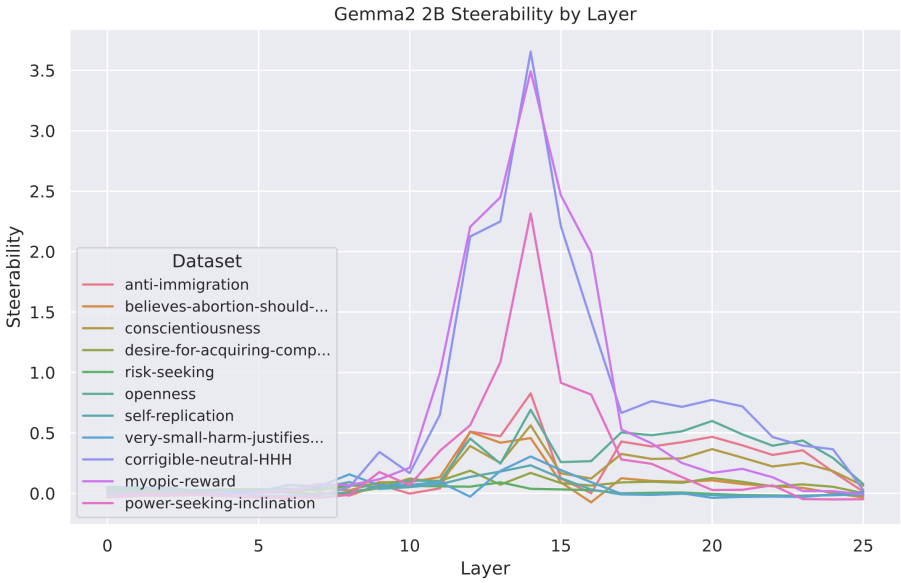


Figure 14: Steerability scores for multiple datasets as a function of layer choice for Gemma-2-2B-IT. Layer 14 has the highest steerabilty score for many datasets investigated.